

Name: Calvin, Andrew, Dylan

Date: 3/11/2024

Class: Engineering Period 3

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

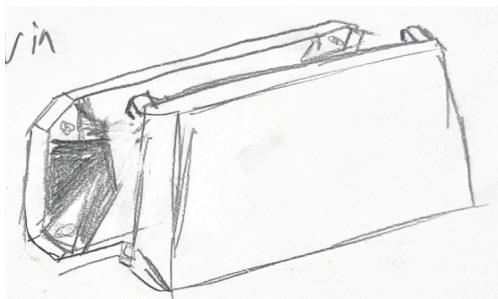
Brainstorming session on 2/8/24 - collection of ideas

Our team individually sketched solutions for the camera lens cover and then collaborated using the “chiming in with friends protocol” where we each got to hear fair criticism on our designs along with possible solutions

Design Idea #1

Date: 2/8/24

Description: This design is a hard plastic cover that snaps onto the back of the box around the lens. It does this by flexing slightly and fitting small knobs into holes that are located in the back of the box. This is shown in the image below

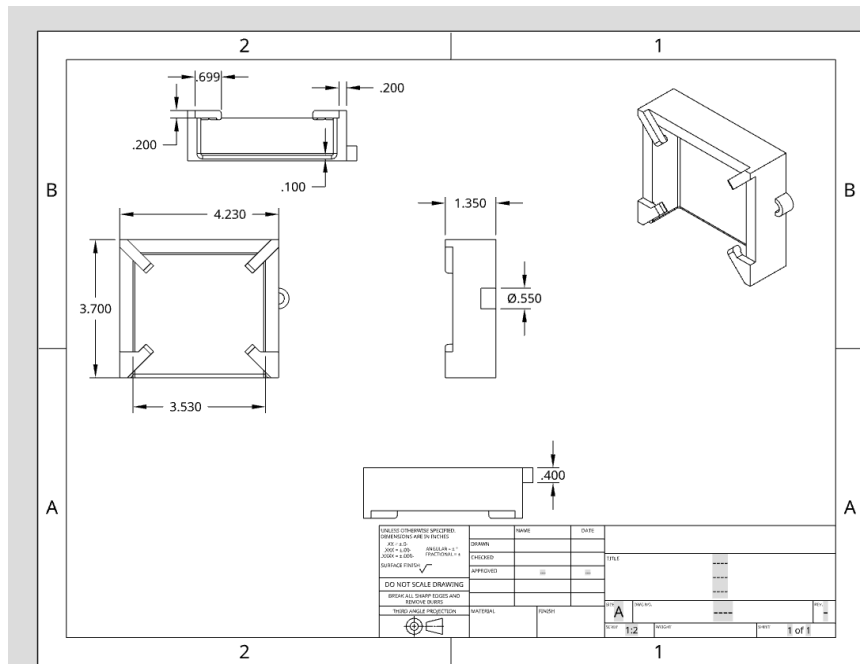


This was the original drawing for the idea when we were brainstorming ideas.

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Dimensions: 3.7x4.23x1.35 in (temporary, may retake)

Material: ABS plastic

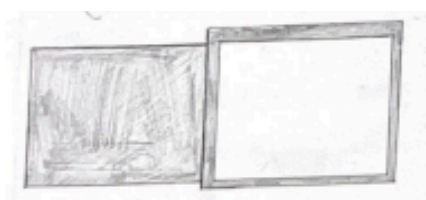
This is the CAD drawing of this idea. We realized that we needed to create a slot on one side of the box for it to actually be able to fit onto the camera.

Design Idea #3

Date: 2/8/24

Description: This design of the lens cover attaches very similarly to the box around the camera just as before, however now instead of a simple wall that goes in front of the lens to protect it, we are now using a sliding cover. This means that you wouldn't have to take it on and off all the time, but simply slide the cover over when filming, and lids it back when done.

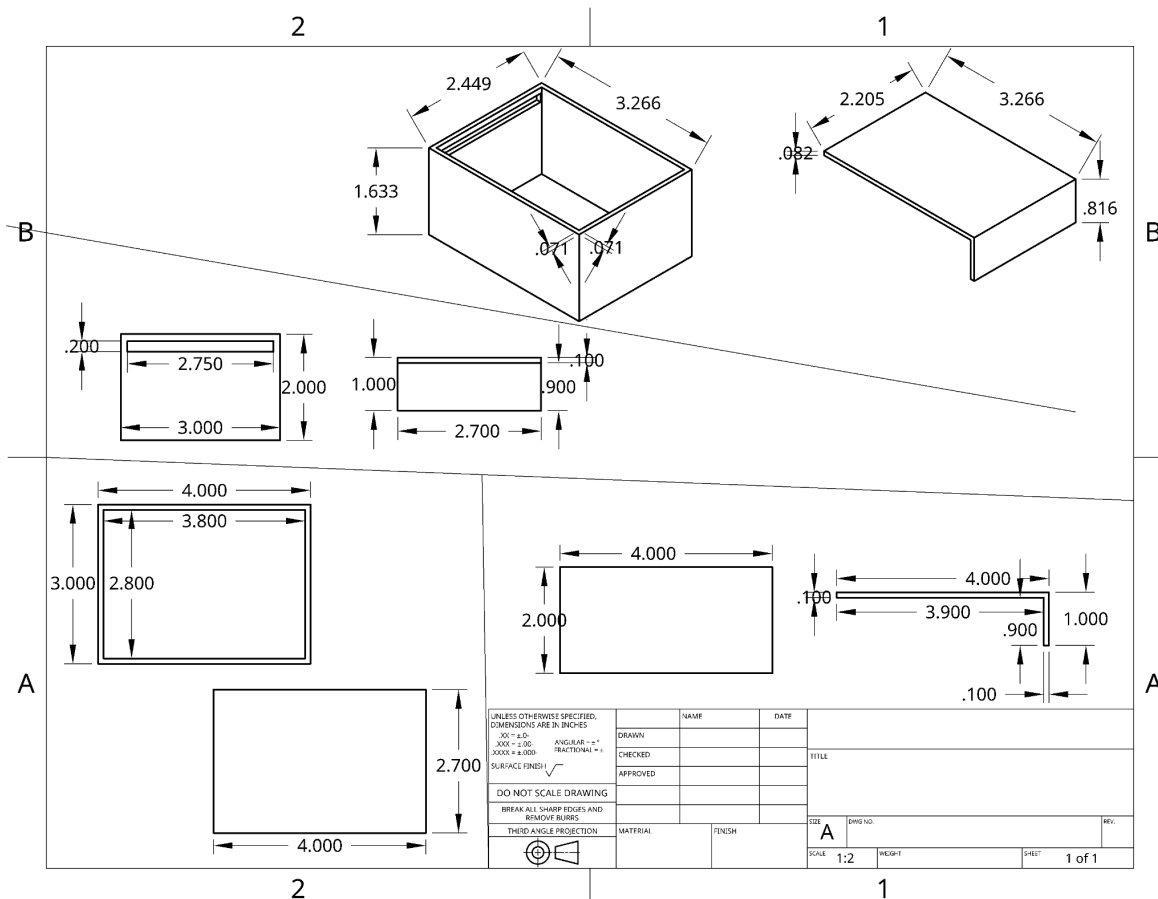
This is a simple drawing of the idea when we were brainstorming for the product.



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Dimensions: 3x4x2 in (temporary, may retake)

Material: ABS plastic

This CAD drawing includes both the box for the camera lens and the sliding door. This design lacked some details that we changed in a later design when revising this.

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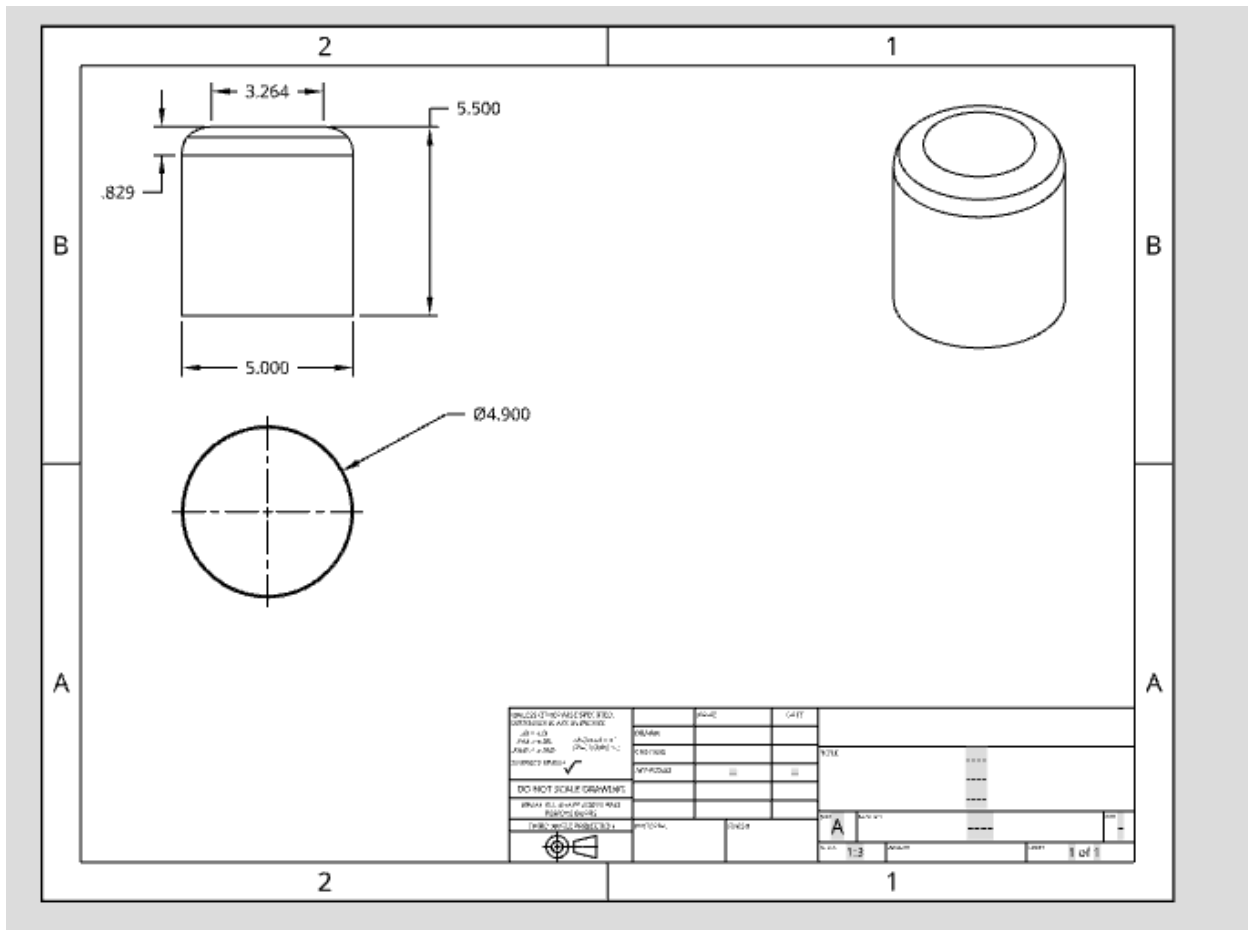
Description: This cover is not rigid, and is instead made of cloth or a similar material. It is flexible and is simply a bag with a drawstring on one end. The bag will be able to fit around a variety of lenses by stretching to match its shape.



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Dimensions: 5.5x4.9x4.9 in (temporary, may retake)

Material: cloth, string

Design Idea Comparison and Selection (D1, D2)

	weight*	Option 1	Option 2	Option 3
Protection of lens	1	1	1	2
Staying securely on the camera	2	2	2	4

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Staying attached to the camera	3	4	5	3
Easily removable	4	5	3	1
Durability	5	3	4	5
score				

*1 is most, 5 is least

Plan of Action (D3)

Week 1	Week 2	Week 3	Week 4
-Develop CAD sketches in OnShape -Take what was planned on paper before and recreate that in OnShape -Meanwhile, take note of dimensions and other important information that can be used to recreate the design later if necessary	-Configure 3D Printer and Prusa settings -Transferring the sketches from Prusa slicer to the 3D Printer -Attempting to 3D Print the sketches with a 3D Printer in the connecting room	-Testing the sketches -Seeing how much they meet the criteria -Calculating which design has the best score (criteria met * weight of said criteria)	-Continue to refine designs with the new knowledge we gained -Plan not known very much past this point

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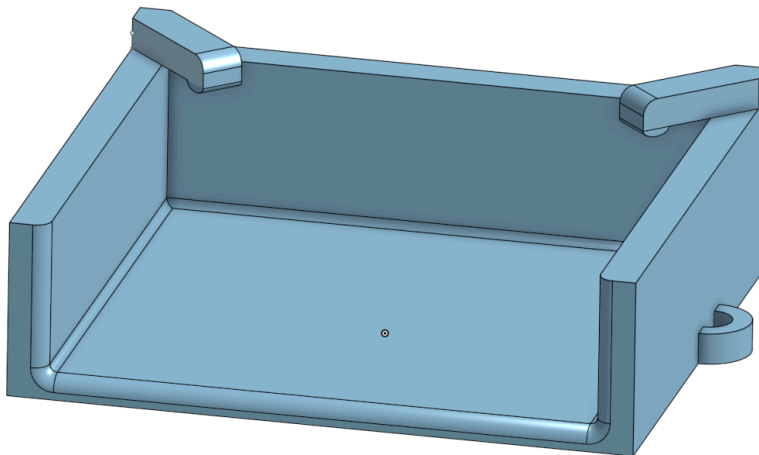
Design Blueprint (D4)

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

Date: 3/27/2024

Sketch (latest version):



Manufacturing Methods:

Step 1: Create CAD design of lens cover

Step 2: Export file as an STL

Step 3: Import STL file into slicing software

Step 4: Print out the product and test if it works

Item	Cost each	# needed	Total
3D Printer	\$230	1	\$230

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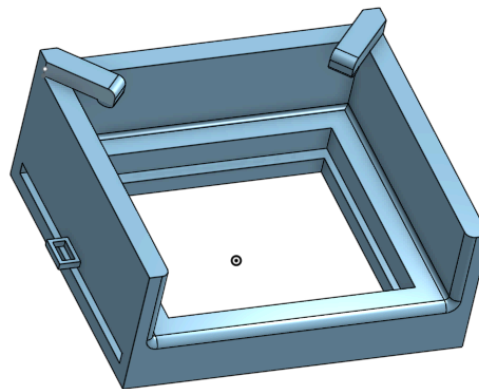
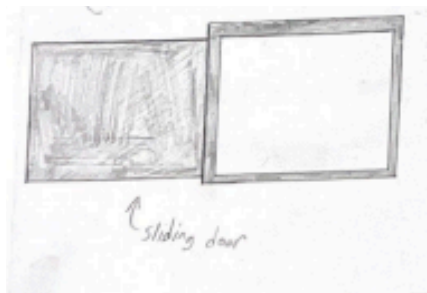
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Filament	\$24.99 per kg	118g	~\$2.95 of spool used per cover, a full spool can make 8 covers
Total Cost			\$232.95

Design Blueprint #2 : Sliding door cover

Date: 3/12/2024

Sketch:



Manufacturing Methods:

Manufacturing Methods:

Step 1: Create CAD design of lens cover

Step 2: Export file as an STL

Step 3: Import STL file into slicing software

Step 4: Print out the frame

Step 5: Print out the sliding door separately, and test

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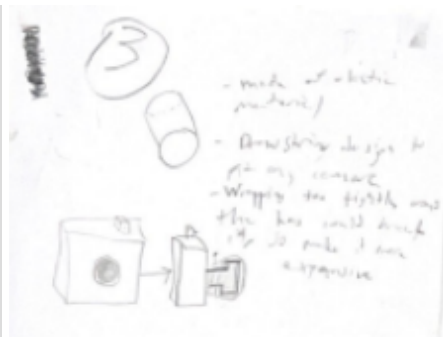
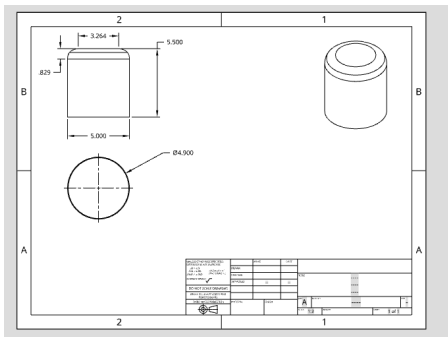
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Supply List and Costs:

Item	Cost each	# needed	Total
3D Printer	\$230	1	\$230
Filament	\$24.99 per kg	37.83g	~\$0.95 per cover, one spool can make about 26 covers
Total Cost			\$230.95

Bag

Date: 3/12/2024



Sketch:

Manufacturing Methods:

Step 1:Sew up bag

Step 2:insert string

Step 3:put string plunger on string

Step 4: Test

Supply List and Costs:

Item	Cost each	# needed	Total
cloth	\$0.80	1 sq ft	\$5.85
Elastic string	\$2.00	1 unit	\$2.00

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String plunger	\$0.60	1 unit	\$1.20
Total Cost	\$3.40	3	\$9.05